Appendix C: IEEE 830 Template

Software Requirements Specification

for

«Kalliopi's Project»

Version «# matching the last line of the versions table»

Prepared by «KalliopiKydonaki»

«organisation»

«11.03.2026»

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Table of Contents

«Table of Contents»

Revision History

Name	Date	Reason for Changes	Version
«KalliopiKydonaki»	«11.03»	«"Initial version" or actual reason»	«Version1»
	25.03	Put inside diagram	Version 2
	29.03	Put diagram	Version 3
	12.04	Diagram	Version 4

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1. Introduction

1.1. Purpose

This document specifies the requirements for the E-Shop system.

1.2. Document Conventions

«Describe any standards or typographical conventions that were followed when writing this specification, such as fonts or highlighting that have special significance. For example, state whether priorities for higher-level requirements are assumed to be inherited by detailed requirements, or whether every requirement statement is to have its own priority.»

1.3. Intended Audience and Reading Suggestions

Developers, Project Managers

1.4. Product Scope

The E-shop system that I want to built, is a web-based application designed to facilitate online retail. It enables customers to browse a digital product catalog, manage virtual shopping cart and complete secure purchases online. The system aims to replace manual shopping processes with a 24/7 automated platform, providing a seamless interface for users and an efficient management dashboard for administrators to track inventory and orders

1.5. References

List any other documents or Web addresses to which this document refers or was based on. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader could access a copy of each reference, including title, author, version number, date, and source or location. Include the assignment document in the references.

2. Overall Description

2.1. Product Perspective

«Describe the context and origin of the product being specified in this specification. For example, state whether this product is a follow-on member of a product family, a replacement for certain existing systems, or a new, self-contained product.»

2.2. Product Features

«Summarize the major features the product must perform or must let the user perform. Details will be provided in Section 3, so only a bullet list is needed here. Organize the features to make them understandable to any reader of the specification. A picture of the major groups of related requirements and how they relate, such as a top level data flow diagram or object class diagram, is often effective.»

- «Thing 1»
- «More things as required»

2.3. User Classes and Characteristics

«Identify the various user classes (intended target for the product) that you anticipate will use this product. User classes may be differentiated based on frequency of use, subset of product functions used, technical expertise, security or privilege levels, educational level, or experience. Describe the pertinent characteristics of each user class. Certain requirements may pertain only to certain user classes. Distinguish the most important user classes for this product from those who are less important to satisfy.»

2.4. Operating Environment

«Describe the environment in which the software will operate, including the hardware platform, operating system and versions, and any other software components or applications with which it must peacefully coexist.»

2.5. Design and Implementation Constraints

«Describe any items or issues that will limit the options available to the developers. These might include: corporate or regulatory policies; hardware limitations (timing requirements, memory requirements); interfaces to other applications; specific technologies, tools, and databases to be used; parallel operations; language requirements; communications protocols; security considerations; design conventions or programming standards (for example, if the customer's organization will be responsible for maintaining the delivered software).»

2.6. User Documentation

«List the user documentation components (such as user manuals, on-line help, and tutorials) that will be delivered along with the software. Identify any known user documentation delivery formats or standards.»

2.7. Assumptions and Dependencies

«List any assumed factors (as opposed to known facts) that could affect the requirements stated in the specification. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project, unless they are already documented elsewhere (for example, in the vision and scope document or the project plan).»

3. System Features

«The purpose here is to give a brief introduction to the system features, so that §4 has something to reference. Then, the full explanation of the system features is given in §5, likely referencing §4. The numbering in §3 and §5 should match.»

3.1. «Feature Name»

«Brief description of the feature, generally 1-3 sentences»

3.«N». «Additional feature blocks as needed»

4. External Interface Requirements

4.1. User Interfaces Overview

«Describe the high level functionality of the system from the user's perspective. Explain how the user should be able to use your system to complete all the expected features and the feedback information that will be displayed for the user (for each screen the user will see). For example, specify whether it is GUI or text based interface and how at high level it would work. Discuss at high level how user will interact with the game such as clicking buttons, typing in some selection character, etc.»

«Describe the minimum requirements for the interface. This may include some sample screen images whether for text or GUI approach, any GUI standards or product family style guides that are to be followed, screen layout constraints, for GUI standard buttons and functions (e.g., help) that must appear on every screen, keyboard shortcuts, error message display standards, and so on»

4.2. Hardware Interfaces

«Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware, and communication protocols to be used.»

4.3. Software Interfaces

«Describe the connections between this product and other specific software components (name and version), including databases, interpreters, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Identify data that will be shared across software

components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.»

5. System Features/Modules

«This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product.»

5.1. «System Feature Name»

5.1.1 Description and Priority

«Provide a short description of the feature. Give its priority, too.»

5.1.2 Stimulus/Response Sequences

«”Stimulus” or “Condition”»: «Describe the stimulus or condition»

Response: «Describe the response»

«Additional Stimulus/Response pairs as needed»

5.1.3 Functional Requirements

«Itemize the detailed functional requirements associated with this feature. These are the software capabilities that must be present in order for the user to carry out the services provided by the feature. Include how the product should respond to anticipated error conditions or invalid inputs. Requirements should be concise, complete, unambiguous, verifiable, and necessary. Each requirement should be uniquely identified with a sequence number or a meaningful tag of some kind.»

REQ-1.1: «Some sentence goes here, in “shall” form, usually something like this: “Upon <some event or condition>, the <system, or the name of a module> shall <perform some action>.” Note that the entire requirement must be stated in the requirements area. Everything else is just a lead-up to this. There should be one such entry per S/R pair.»

REQ-1.«N»: «Additional S/R-related requirements as needed.»

REQ-1.«N»: «List any additional requirements needed in “Shall” form, but not necessarily in “Upon/Shall” form.»

REQ-1.«N»: «Additional requirements as needed.»

5.«N». «Additional feature blocks»

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: «Any performance-related issue»

NF-1.«N»: «Any performance-related issue»

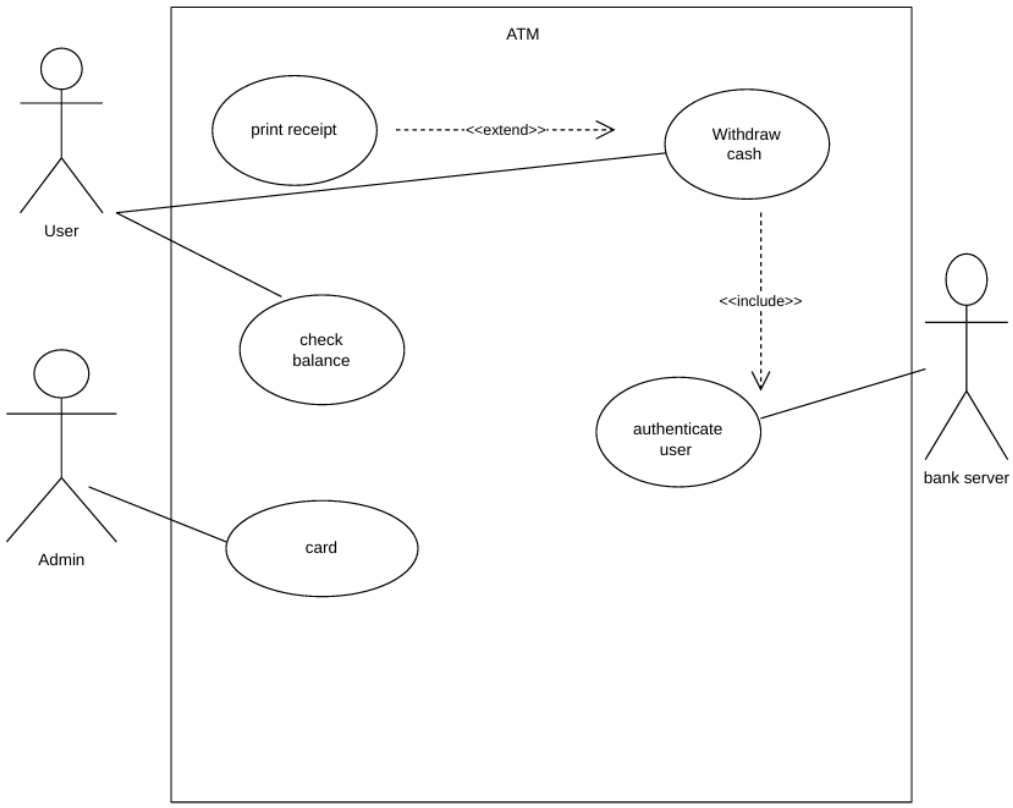
6.2. Security

NF-2.1: «Any security-related issue»

NF-2.«N»: «Any security-related issue»

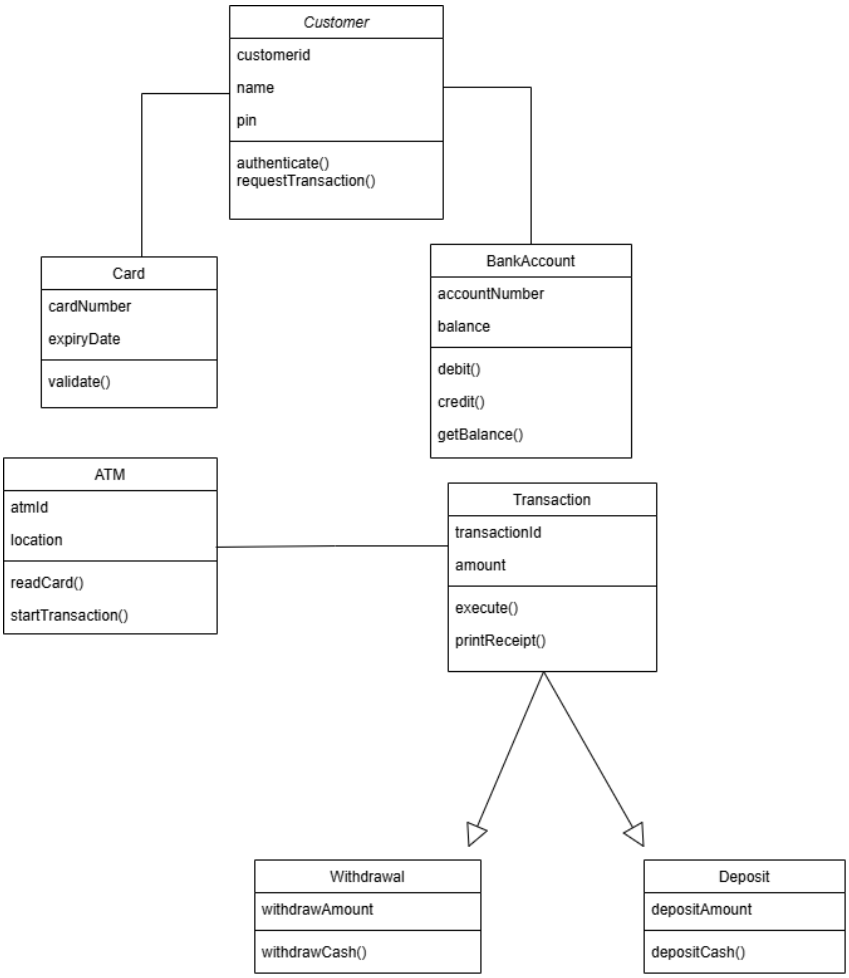
6.«N». «Any additional “nonfunctional” blocks»

Diagram :

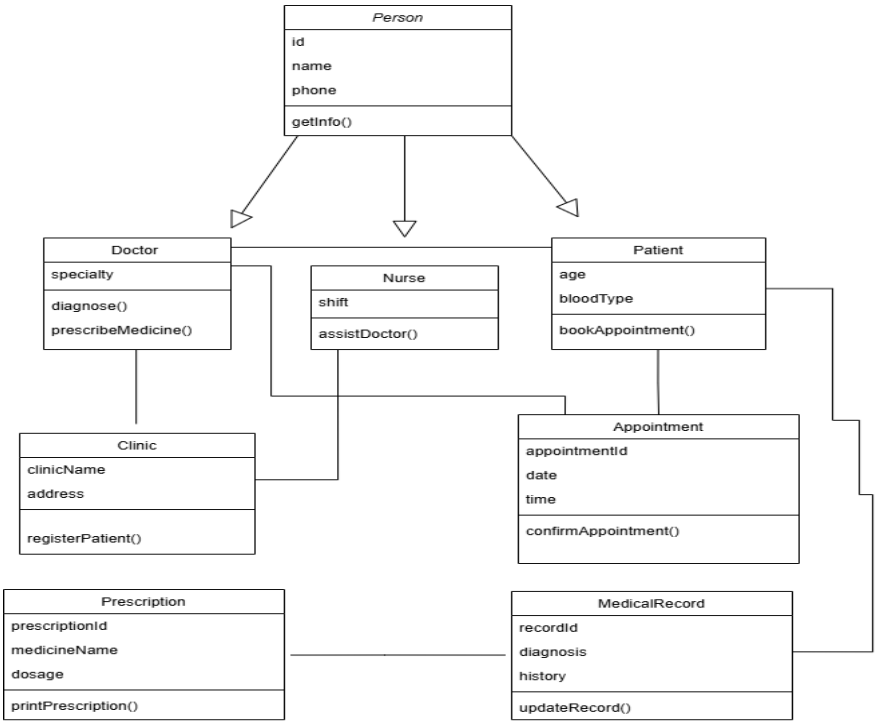


This diagram shows the main functions of an ATM system. The user can withdraw cash and check balance, while the bank server is used for user authentication. It also shows that printing a receipt is an optional step after cash withdrawal.

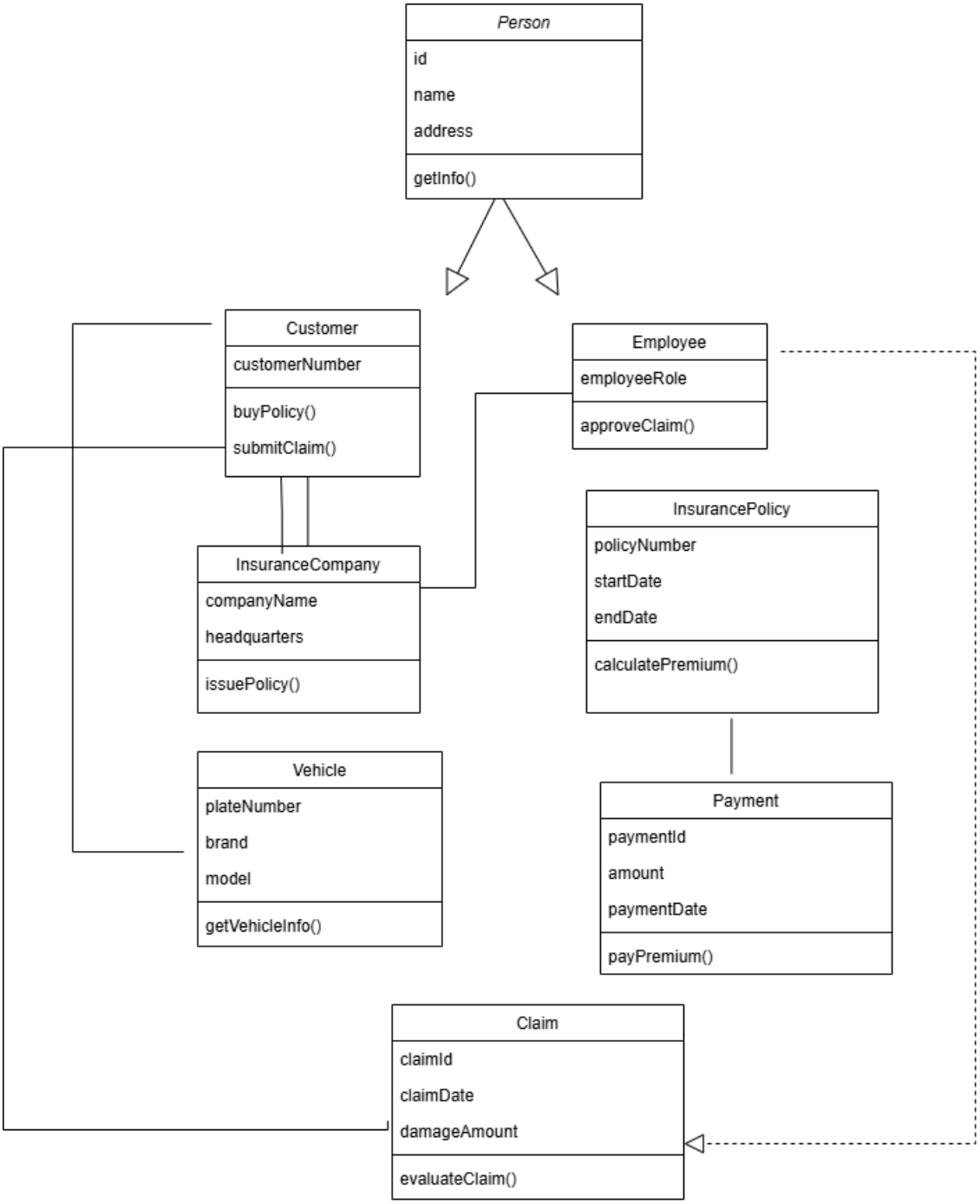
System 1 :



System 2 :

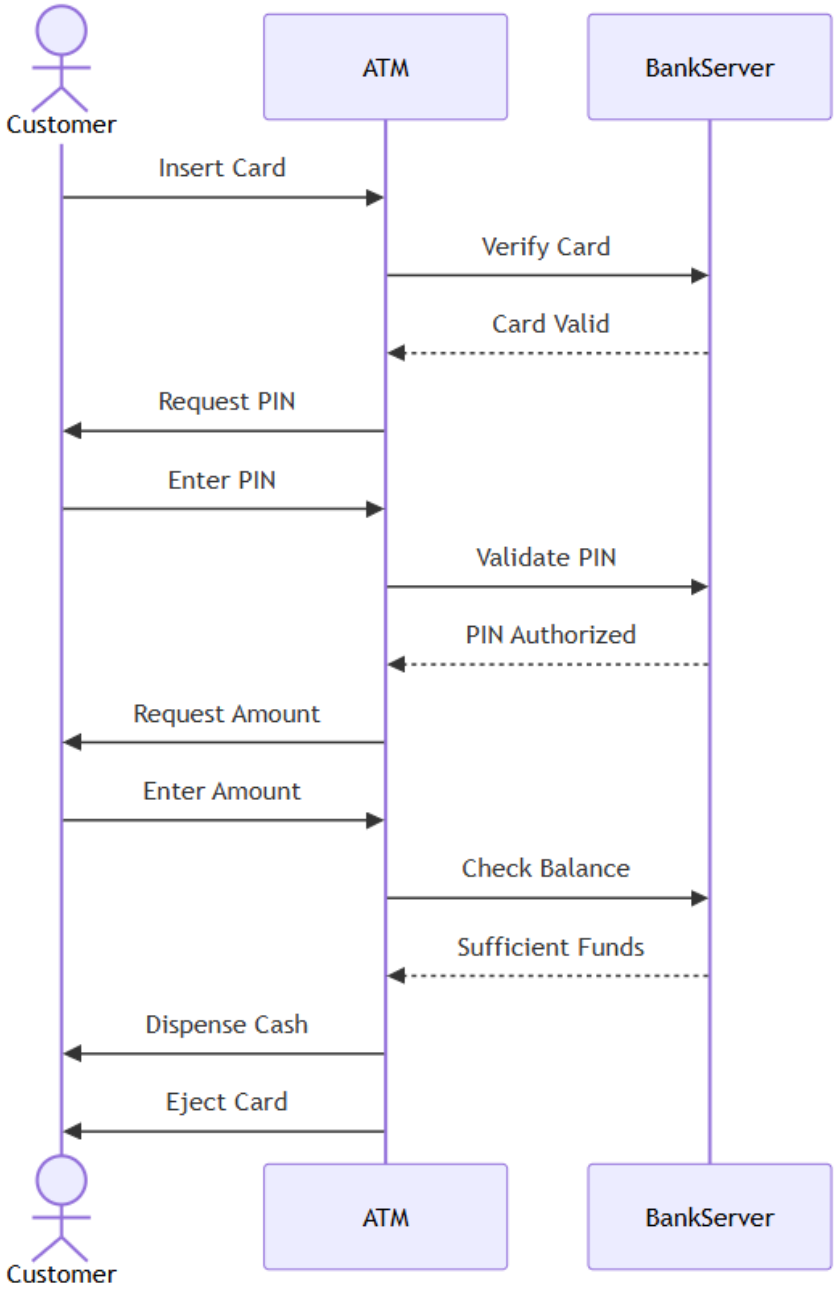

Person
id
name
phone
getInfo()
Doctor
specialty
diagnose()
prescribeMedicine()
Nurse
shift
assistDoctor()
Patient
age
bloodType
bookAppointment()
Clinic
clinicName
address
registerPatient()
Appointment
appointmentId
date
time
confirmAppointment()
Prescription
prescriptionId
medicineName
dosage
printPrescription()
MedicalRecord
recordId
diagnosis
history
updateRecord()

System 3 :



This assignment presents three UML class diagrams: ATM System, Medical Clinic System, and Car Insurance System. Each diagram includes classes, attributes, methods, and relationships between classes. The ATM diagram focuses on customer transactions, the Medical Clinic diagram focuses on staff and patient records, and the Car Insurance diagram focuses on the relationship between the customer and the insurance company. These diagrams help describe the structure and organization of each system.

Task 4 :


Customer
ATM
BankServer
Insert Card
Verify Card
Card Valid
Request PIN
Enter PIN
Validate PIN
PIN Authorized
Request Amount
Enter Amount
Check Balance
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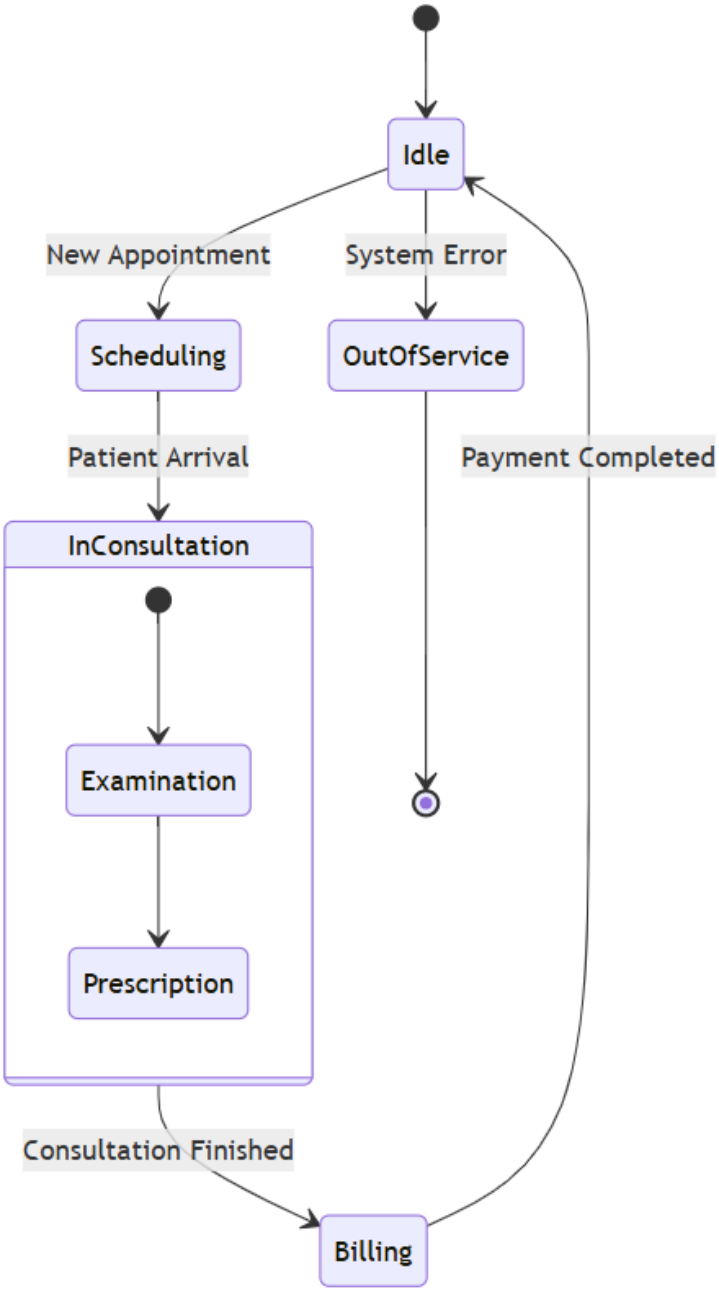
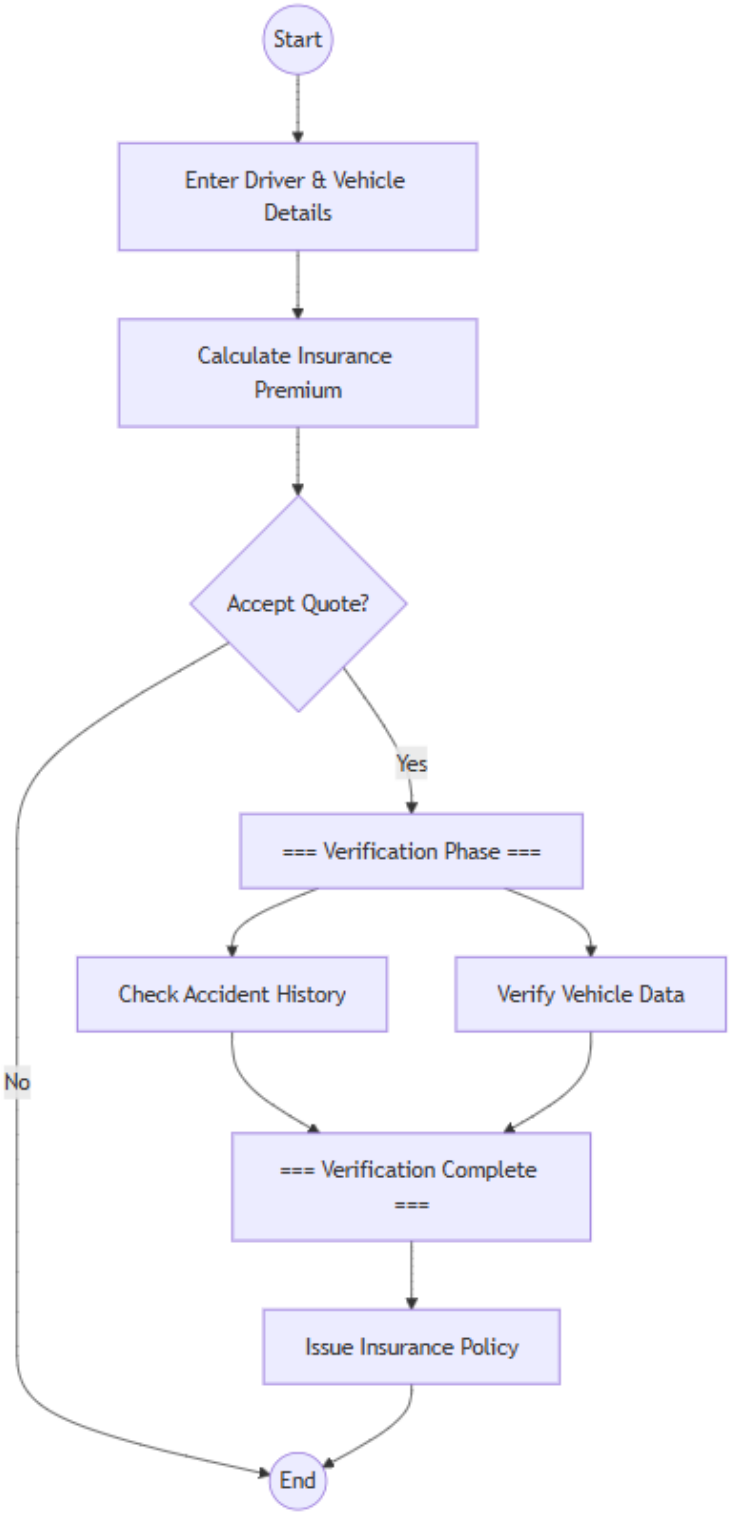
Eject Card
Customer
ATM
BankServer


Idle
New Appointment
System Error
Scheduling
OutOfService
Patient Arrival
Payment Completed
InConsultation
Examination
Prescription
Consultation Finished
Billing


Start
Enter Driver & Vehicle Details
Calculate Insurance Premium
Accept Quote?
Yes
No
=== Verification Phase ===
Check Accident History
Verify Vehicle Data
=== Verification Complete ===
Issue Insurance Policy
End